11. AI Across Civil and Architectural Engineering

The big picture

Week 11 · CE/ARCE 2XX AI Literacy for Civil and Architectural Engineering



COLLEGE OF ENGINEERING
Civil & Architectural Engineering & Mechanics

Agenda

Today's plan

1. Quiz on Week 10 (AI agents)
2. Where AI fits in the life cycle of an asset
3. A tour by sub-discipline: seven examples
4. Digital twins, and combining physics with data
5. Why CE and ARCE data is hard
6. Case-study jigsaw
7. Lab 11 preview and your project proposal

Recap

Last week in 60 seconds: AI agents

1 Model + tools + memory + loop

Observe → think → act → observe, until done or a step limit stops it.

2 Tool calling

The model writes a JSON request; your code checks it and runs the tool.

3 Risks

Compounding errors, wrong inputs, prompt injection, unsafe actions, cost.

4 Guardrails

Least privilege, human approval, limits, logs, tests — and use the simplest tool that works.

Learning objectives

By the end of today, you can…

- ✓ Place AI applications on the life cycle: plan, design, build, operate, maintain, renew
- ✓ For an application, name its input data, AI task type, and output
- ✓ Explain digital twins, surrogate models, and physics-informed ML
- ✓ Explain why CE/ARCE data is hard — and why accuracy misleads for rare failures
- ✓ Analyze a case study and judge whether AI fits the task

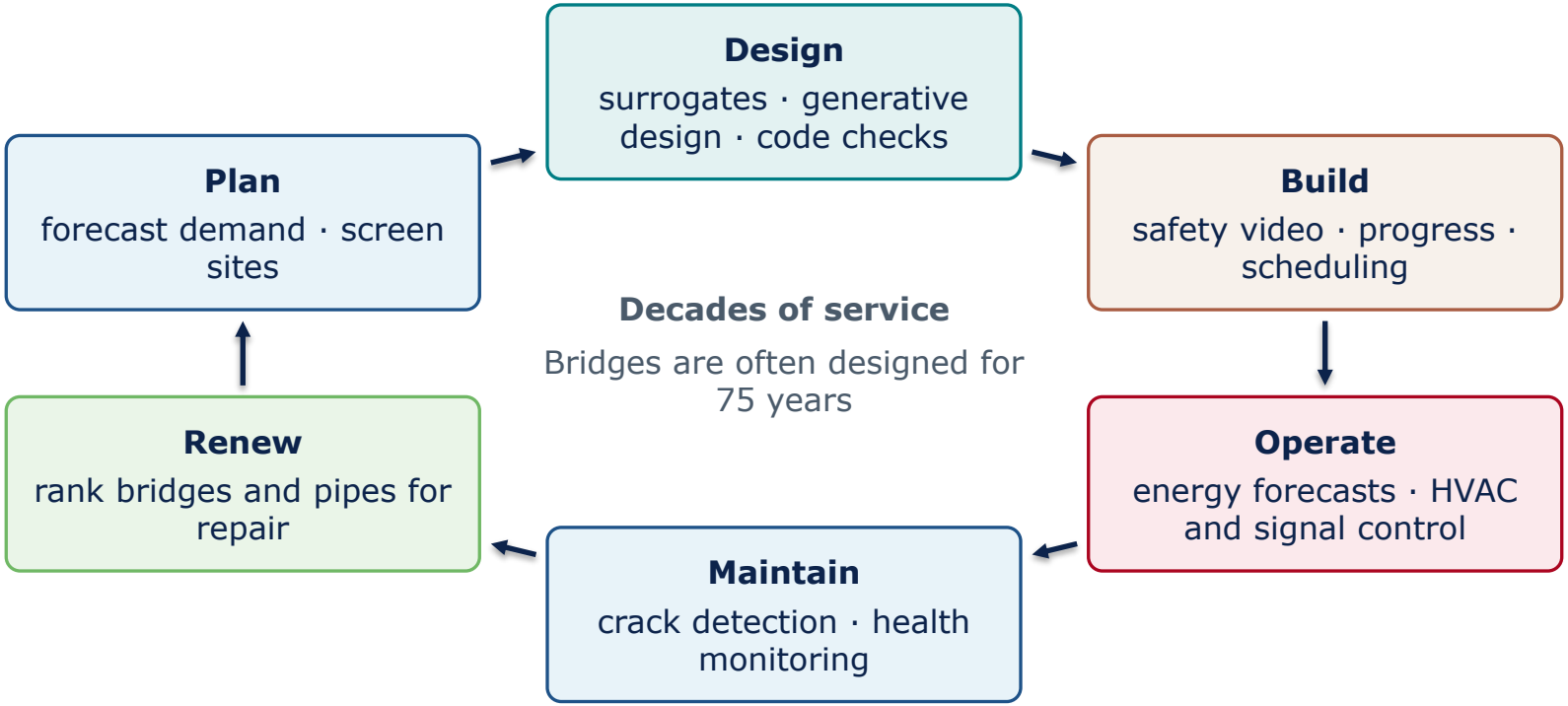
01

The life cycle

AI can help in every phase of an asset's life

The life cycle

AI shows up in every phase of an asset's life



The life cycle

Six task types cover most engineering AI

WEEKS 3–5

Regression

Predict a number. Heating load (kWh/m²) from building shape.

WEEKS 4–6

Classification

Pick a category. Cracked or not; poor deck or not.

WEEK 6

Detection

Find where things are in an image: hard hats, potholes, spalls.

WEEK 6

Forecasting

Predict future values in a time series: traffic next hour, river level tonight.

WEEK 1 MAP

Optimization

Choose the best option under limits: repairs within a budget, HVAC setpoints.

WEEKS 7–8

Generation

Create new content: reports, code, images, design options.

02

A tour by sub-discipline

Seven examples: input data → AI task → output

Structural

Health monitoring learns “normal” and flags the unusual

INPUT DATA

Acceleration and strain time series from sensors on a bridge or tall building, plus air temperature

↓

AI TASK · ANOMALY DETECTION

Learn the normal vibration pattern (e.g., natural frequencies) from a healthy period; flag departures

↓

OUTPUT

An alert with the time and sensor location → an engineer inspects

Who uses it

Owners of long-span bridges, tall buildings, and stadiums; researchers

Watch out

Temperature and traffic also shift frequencies → false alarms. Few or no “damaged” examples.

Also in this field

Crack and spall detection in photos (Week 6); fast surrogate models for design (later today)

Geotechnical · pre-reading

Mapping where slopes are likely to fail

INPUT DATA

Map layers per grid cell: slope angle, geology, soil, rainfall, land cover, distance to roads and streams — plus past landslides

↓

AI TASK · CLASSIFICATION

Landslide or not, for every cell, as a probability — often logistic regression or a random forest (many decision trees that vote)

↓

OUTPUT

A susceptibility map (low → very high) for routing roads and pipelines

Who uses it

Geological surveys, state departments of transportation (DOTs), and planners

Watch out

Inventories miss small or old slides. Nearby cells look alike, so test on a separate region.

Also in this field

Soil classification from cone penetration tests (a cone pushed into the ground measures resistance)

Transportation

Forecasting how many vehicles are coming

INPUT DATA

Hourly counts from road sensors or cameras, time of day, day of week, holidays, weather (like Lab 06's I-94 data)

↓

AI TASK · FORECASTING

A sliding window of recent hours predicts the next hour's volume (vehicles/h)

↓

OUTPUT

Predicted vehicles/h per road segment → signal timing, ramp meters, work-zone schedules

Who uses it

State DOTs and city traffic operations centers

Watch out

Crashes, events, and new roads break past patterns. Always beat a naive baseline.

Also in this field

Pavement cracks and potholes from vehicle cameras (detection); adaptive signals; automated vehicles

Forecasting river levels hours ahead

INPUT DATA

Rain gauges and radar, upstream stream gauges (level and flow), soil moisture, weather forecasts

↓

AI TASK · FORECASTING

Predict water level or flow 1–24 h ahead — often an LSTM (Week 6) or a hybrid with a hydrologic model

↓

OUTPUT

A forecast hydrograph (level over time) + a warning when a threshold is crossed → close crossings

Who uses it

Flood control districts, forecasters, and water utilities

Watch out

The biggest floods are the rarest in the record. Gauges can fail in the very storms you care about.

Also in this field

Leak detection from pressure and acoustic sensors in water mains; water quality; stormwater

Construction

Spotting missing hard hats in site video

INPUT DATA

Frames from site cameras or drones, labeled with boxes around workers, hard hats, and vests

↓

AI TASK · OBJECT DETECTION

A CNN-based detector finds each worker and checks for PPE (personal protective equipment: hard hat, vest)

↓

OUTPUT

Boxes on the image + an alert to the safety manager; weekly trend reports

Who uses it

General contractors and safety managers

Watch out

Worker privacy. Misses at night, in rain, or when people are partly hidden. Not a replacement for a competent person.

Also in this field

Progress tracking (site scans vs the BIM model); schedule risk; cost estimating from past bids (regression)

Building systems (ARCE)

Predict building energy use, then control it

INPUT DATA

Geometry and envelope, schedules, weather, smart-meter and building-automation trend data

↓

AI TASK · REGRESSION → OPTIMIZATION

Predict heating, cooling, or hourly energy (as in Lab 05); then search setpoints that cut energy within comfort limits

↓

OUTPUT

Energy forecasts, recommended setpoints and schedules, fault alerts (e.g., a stuck damper)

Who uses it

Energy engineers, facility managers, and early-stage designers

Watch out

Models trained on one climate or building type transfer poorly (Phoenix is not Belgium). Enforce comfort and safety limits.

Also in this field

Fault detection; occupant comfort; daylighting; generative design of layouts and façades; BIM queries

Asset management

Which bridges need attention first?

INPUT DATA

Inventory and inspection records: age, material, design type, traffic, truck share, past ratings (the National Bridge Inventory)

↓

AI TASK · CLASSIFICATION + OPTIMIZATION

Predict which decks will be rated poor; then choose repairs that cut the most risk within the budget

↓

OUTPUT

A risk-ranked list of bridges for inspection or repair → engineers decide

Who uses it

State DOTs, counties, and cities

Watch out

Poor decks are rare — about 1% in Arizona. Ratings are human judgments. A model never replaces an inspection.

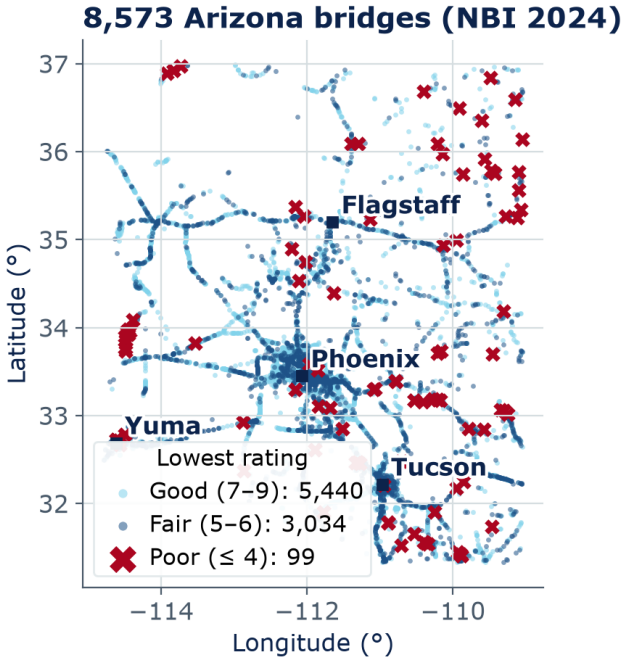
Also in this field

Deterioration models predict condition years ahead; pavement and pipe management work the same way

Asset management

8,573 Arizona bridges: most are fair or good

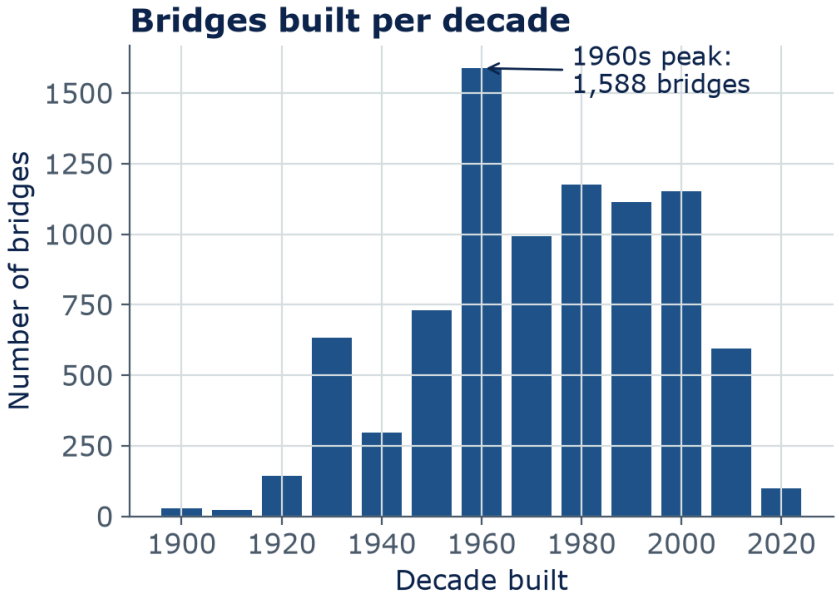
- All 8,573 structures, by lowest component rating: **good** 5,440 (63%) · **fair** 3,034 (35%) · **poor bridges 99 (1.2%)** (any component ≤ 4)
- Poor bridges are not spread evenly: **44 of the 99** are owned by tribes or the Bureau of Indian Affairs (BIA) — 11% of those 388 bridges
  - Their repair is funded through a separate federal program (23 CFR 661): an infrastructure-investment question
- 56% of the rows are culverts, which get a culvert rating instead of a deck rating

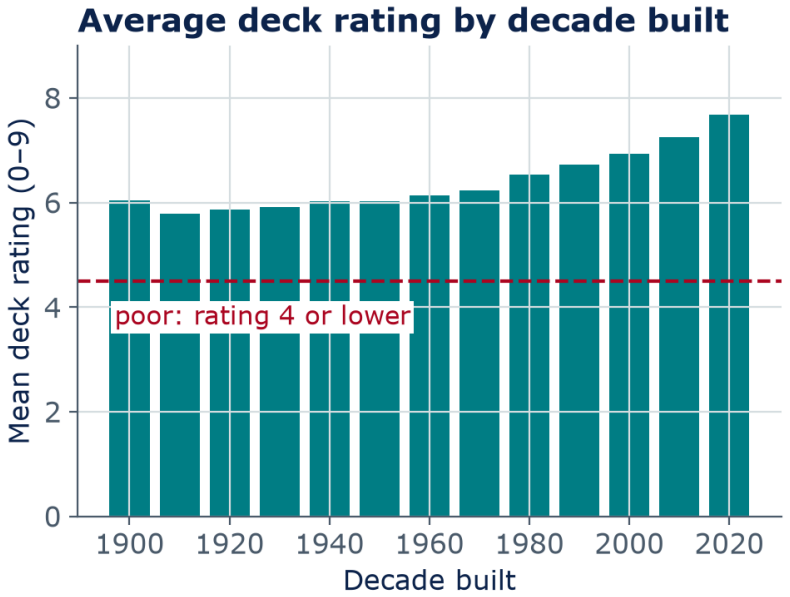


Source: FHWA National Bridge Inventory, Arizona 2024

Asset management

Older Arizona bridges have lower deck ratings





Left: all 8,573 bridges. Right: the 3,762 bridges with a deck rating. Source: FHWA NBI, Arizona 2024.

Concept check

Which task type is it?

A DOT has a risk score for every bridge and must choose which repairs to fund this year to cut the most risk within a fixed budget. Which task type is the choosing step?

A Regression

B Classification

C Optimization

D Generation

Concept check

Which task type is it? — answer

A DOT has a risk score for every bridge and must choose which repairs to fund this year to cut the most risk within a fixed budget. Which task type is the choosing step?

A Regression

B Classification

C Optimization

D Generation

Choosing the best set of actions under a limit (the budget) is optimization; the risk scores came from classification.

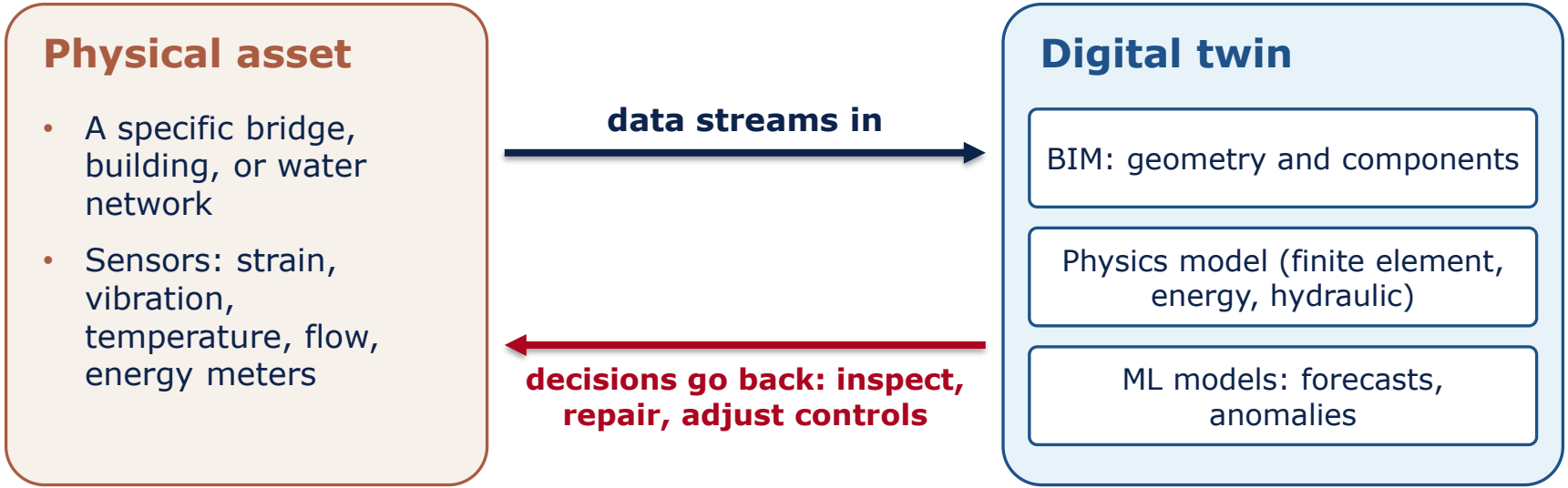
03

Digital twins, physics + data

Models that stay connected to the real asset

Digital twins

A digital twin stays connected to the real asset

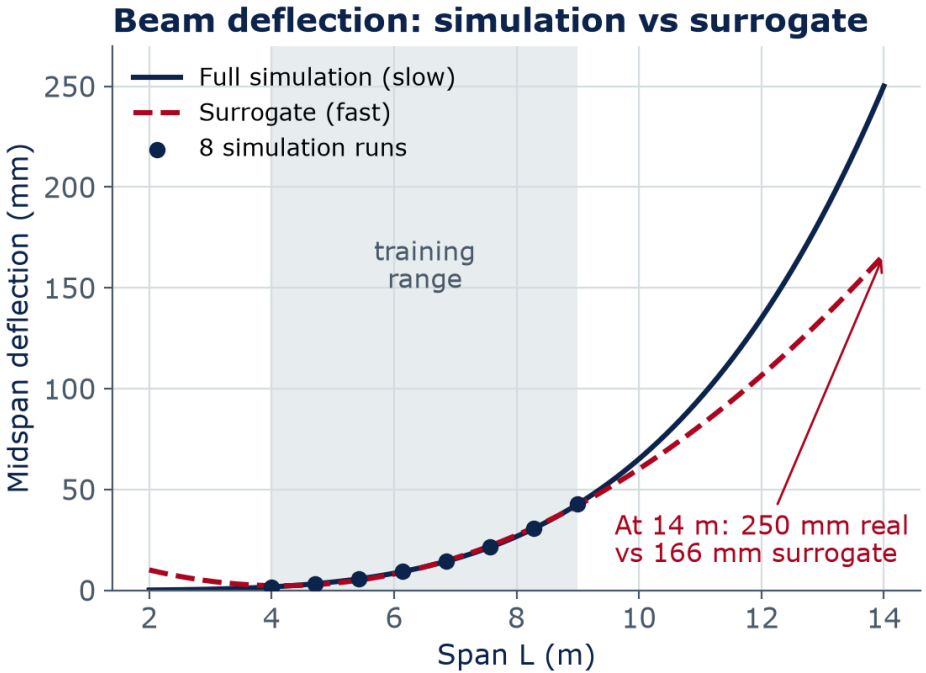


Digital model	Digital shadow	**Digital twin**
no automatic data link	data flows in, one way	**data in, decisions out**

Physics + data

Surrogate models: fast, but only inside their range

- A **surrogate** is a fast ML model trained on runs of a slow physics model (e.g., finite-element analysis)
- Explore thousands of design options in seconds — then check the best few with the full model
- Inside the training range it can be very close…
- …outside, it can be **badly wrong**: here it under-predicts deflection by a third at 14 m — the unsafe direction
  - This beam fails L/360 at 14 m anyway (250 mm ≈ L/56); a surrogate only screens options



Physics + data

Physics-informed ML adds a physics penalty

$$J_{total} = J_{data} + \lambda J_{physics}$$

J_data	error against measurements — the cost from Week 3
J_physics	how badly predictions break physics (e.g., mass balance)
λ	a weight you choose: how much to trust physics vs data

Example: a water-network model penalized whenever flow into a junction ≠ flow out

04

Why CE and ARCE data is hard

Small, rare, long-lived, and safety-critical

Why the data is hard

Six reasons engineering data is harder than web data

1 Small data

Hundreds of bridges or test cylinders, not millions of photos. Every sample costs money.

2 Rare failures

Collapses and big floods are rare, so models see few examples of what matters most.

3 Long lifespans

Assets last 50–100 years; climate, loads, codes, and sensors change meanwhile.

4 Safety-critical

A wrong answer can hurt people. “Usually right” is not good enough.

5 Codes and liability

Designs must meet codes, and a licensed engineer is responsible for the result.

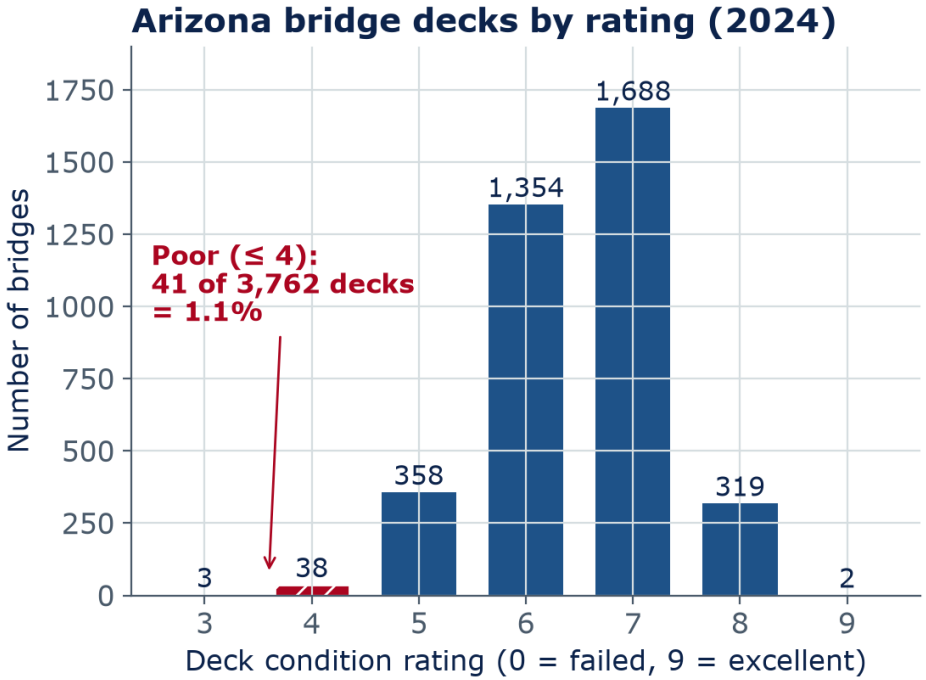
6 Messy records

In our bridge file: 13 rebuild years coded 9999; a 9.8 m culvert listed as 990.6 m wide.

Why the data is hard

Rare failures make accuracy misleading

- Of 3,762 rated Arizona decks, only **41** are poor (≤ 4): 1.1%
- A "model" that always says "not poor" is **98.9% accurate**…
- …and finds **0 of the 41** poor decks
- **Recall** = poor decks found ÷ all poor decks. For rare failures, report recall, not just accuracy



Concept check

Recall or precision?

A model flags 90 decks as poor: 30 of them really are poor, and it misses the other 11 of the 41 poor decks. What is its recall for poor decks?

A About 33%

B About 73%

C About 98%

D About 1%

Concept check

Recall or precision? — answer

A model flags 90 decks as poor: 30 of them really are poor, and it misses the other 11 of the 41 poor decks. What is its recall for poor decks?

A. About 33%

B. About 73%

C. About 98%

D. About 1%

Recall = 30 ÷ 41 ≈ 73%. (30 ÷ 90 ≈ 33% is precision: the share of flags that were right.)

05

Case-study jigsaw

Become an expert on one case, then teach your group

Case-study jigsaw

Jigsaw: one case per group, then teach each other

16 min

Jigsaw

- **Expert groups (6 min):** each group takes one case (A–F) and fills in the template below
- **Home groups (8 min):** one expert per case; each teaches their case in 1 minute
- As a home group, pick the case **most** ready to deploy and the **least** ready, with one reason each
- **Share-out (2 min):** two home groups report

Template: input data → AI task type → output and who uses it → success metric and baseline → what could go wrong, and who is accountable?

Case-study jigsaw

Six cases for the expert groups

A · ASSET MANAGEMENT

Bridge deck triage

A state DOT wants to flag decks likely to be rated poor at the next inspection.

B · WATER

Flash-flood warning

A county wants 1-hour flood warnings at low-water crossings from rain and stream gauges.

C · CONSTRUCTION

Hard-hat alerts

A contractor wants camera alerts when someone lacks PPE (hard hat, vest).

D · BUILDING SYSTEMS

Campus energy

Facilities wants next-day electricity forecasts for 20 buildings and HVAC fault alerts.

E · TRANSPORTATION

Street pavement survey

A city wants to rate every street's pavement from city-vehicle dash-camera images.

F · BUILDING CODES

Code-check agent

A design firm wants an LLM agent to check drawings against accessibility rules.

Lab 11

Lab day: end-to-end ML on Arizona bridges

LAB

11

Bring a charged laptop

Open the notebook from the course GitHub page

- Frame the question: which bridge decks are rated poor (≤ 4)?
- Explore 8,573 bridges: map by latitude and longitude, age vs condition
- Build features; compare a baseline (always “not poor”) with logistic regression and a random forest
- Accuracy vs recall for the rare poor class; feature importance
- Write a 5-sentence engineering memo with caveats — your project template

Final project

Proposal due Sunday: a two-page project canvas

Due Sunday 11:59 pm with your team charter · one per team · 10% of the project grade

Problem and users

Who has the problem? What decision does your work support?

Data and its limits

Source, size, labels, license, what is missing.

Method and baseline

Your track, your model, and the simple thing to beat.

Evaluation

Which metrics, and what counts as good enough?

Verification

How you will check results: hand calcs, known answers, tests.

Risks and accountability

Failure modes, bias, who is responsible.

Plan and roles

Who does what; the check-in target; a backup plan.

AI-use plan

Which AI tools you will use, for what, and how you will document them.

Wrap-up

Key takeaways

1. AI appears in every life-cycle phase: plan, design, build, operate, maintain, renew
2. Describe any application as **input data** → **task type** → **output**, and ask who acts on it
3. Digital twins stay linked to the real asset; surrogates and physics-informed ML combine physics and data
4. CE/ARCE data is small, imbalanced, long-lived, and safety-critical — **check recall**, not just accuracy
5. AI ranks, flags, and forecasts; engineers inspect, decide, and stay accountable

Learn more

Readings and resources

- **Google ML Crash Course: accuracy, precision, recall** — required: the metrics behind this week's lab (20 min)
- **FHWA National Bridge Inventory** — required: where our Arizona data comes from (10 min)
- **National Academies (2024): Foundational Research Gaps and Future Directions for Digital Twins** — optional: free PDF; the summary (20 min)
- **Raissi, Perdikaris & Karniadakis (2019): Physics-informed neural networks** — the PINN paper — skim the introduction (optional)